

Statistical and Machine Learning

Dimension reduction – Exercises (Teacher version with solutions)

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Question 1 – PCA: encoding, decoding, and variance

PCA can be derived either by minimising the reconstruction error or by maximising the variance of the projected data.

- a) Explain in your own words what is meant by the “encode” and “decode” steps in PCA, and write down the reconstruction error $\mathcal{L}(\mathbf{W})$.

Solution. Encoding maps $\mathbf{x} \in \mathbb{R}^D$ to a low-dimensional code $\mathbf{z} = \mathbf{W}^\top \mathbf{x} \in \mathbb{R}^L$. Decoding maps the code back to the original space, $\hat{\mathbf{x}} = \mathbf{W}\mathbf{z}$. The reconstruction error (distortion) is

$$\mathcal{L}(\mathbf{W}) = \frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \text{decode}(\text{encode}(\mathbf{x}_n; \mathbf{W}); \mathbf{W})\|_2^2 = \frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \mathbf{W}\mathbf{W}^\top \mathbf{x}_n\|_2^2.$$

- b) What is the optimal choice of $\mathbf{W}$ that minimises this error, and how is it related to the empirical covariance matrix $\hat{\Sigma}$?

Solution. $\hat{\mathbf{W}} = \mathbf{U}_L$, the matrix whose columns are the L eigenvectors of $\hat{\Sigma} = \frac{1}{N} \mathbf{X}_c^\top \mathbf{X}_c$ corresponding to the L largest eigenvalues.

- c) Explain why minimising the reconstruction error is equivalent to maximising the variance of the projected data. Why does this make PCA find the “directions of maximal variance”?

Solution. After substituting the optimal $z_{n1}^* = \mathbf{w}_1^\top \mathbf{x}_n$, the loss becomes $\mathcal{L}(\mathbf{w}_1) = \text{const} - \frac{1}{N} \sum_n z_{n1}^2 = \text{const} - \mathbf{w}_1^\top \hat{\Sigma} \mathbf{w}_1$. Since the data is centered, $\frac{1}{N} \sum_n z_{n1}^2$ is exactly the variance of the projected coordinates $\hat{z}_1$. Hence minimising $\mathcal{L}$ is the same as maximising $\mathbb{V}[\hat{z}_1]$. The maximiser (subject to $\|\mathbf{w}_1\| = 1$) is the eigenvector of $\hat{\Sigma}$ with the largest eigenvalue – i.e. the direction of maximal variance.

Question 2 – Choosing the number of components L

Several criteria were discussed for choosing the latent dimension L in PCA.

- a) What is a scree plot, and how can it be used to choose L ?

Solution. A scree plot shows the eigenvalues λ_j in decreasing order against their index j . Since the reconstruction error for a model with L components is $\mathcal{L}_L = \sum_{j=L+1}^D \lambda_j$, the plot decreases monotonically as L increases. In practice, one looks for an “elbow” – a point where the eigenvalues level off – and chooses L around that point.

- b) Define the “fraction of variance explained” F_L . How does it behave as L increases, and how can it be used to choose L ?

Solution.

$$F_L = \frac{\sum_{j=1}^L \lambda_j}{\sum_{j'=1}^{L^{\max}} \lambda_{j'}}.$$

It is the complement of the (rescaled) reconstruction error, so it increases monotonically from 0 to 1 as L increases. One choice is to pick the smallest L such that F_L exceeds some threshold (e.g. 90% of variance explained).

- c) Briefly describe the idea behind the profile-likelihood method for choosing L . What quantity is being modelled, and what is being optimised?

Solution. The sorted eigenvalues $\lambda_1 \geq \dots \geq \lambda_{L^{\max}}$ are treated as data, and for each candidate L we fit a two-group Gaussian model with shared variance: the first L eigenvalues come from $\mathcal{N}(\mu_1, \sigma^2)$ and the remaining from $\mathcal{N}(\mu_2, \sigma^2)$. The profile log-likelihood $\ell(L)$ is computed using the MLEs $\mu_1(L), \mu_2(L), \sigma^2(L)$, and L^* is chosen as the value of L that maximises $\ell(L)$ – i.e. it automates detection of the “elbow”/change-point in the eigenvalue spectrum.

Question 3 – From PCA to Factor Analysis

Factor analysis (FA) generalises PCA by introducing a probabilistic generative model.

- a) Write down the generative model for FA (the distributions for $\mathbf{z}$ and $\mathbf{x} \mid \mathbf{z}$), and the resulting marginal distribution $p(\mathbf{x})$.

Solution.

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I}), \quad p(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{x} \mid \mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \boldsymbol{\Psi}),$$

and marginalising over $\mathbf{z}$ gives

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{W}\mathbf{W}^\top + \boldsymbol{\Psi}).$$

- b) What role does the matrix $\boldsymbol{\Psi}$ play, and why must it be restricted to be diagonal?

Solution. $\boldsymbol{\Psi}$ is the covariance of the observation noise – it captures the “uniqueness” of each visible variable (variance not explained by the common latent factors). It must be diagonal because otherwise we could set $\mathbf{W} = \mathbf{0}$ and let $\boldsymbol{\Psi}$ model any covariance structure directly, making the latent factors entirely redundant.

- c) Compare the latent representation obtained from FA, $p(\mathbf{z} \mid \mathbf{x})$, with the latent representation obtained from PCA. What is the key conceptual difference?

Solution. In PCA the embedding is a deterministic projection, $\mathbf{z} = \mathbf{W}^\top \mathbf{x}$. In FA, $p(\mathbf{z} \mid \mathbf{x})$ is a full Gaussian posterior distribution with mean $\mathbf{W}^\top \mathbf{C}^{-1}(\mathbf{x} - \boldsymbol{\mu})$ and a non-trivial covariance $\mathbf{I} - \mathbf{W}^\top \mathbf{C}^{-1} \mathbf{W}$. The key difference: FA represents uncertainty about the latent factors, whereas PCA gives only a single point estimate.

Question 4 – Identifiability in Factor Analysis

The parameters of an FA model are not uniquely identifiable.

- a) Show (in words or with the relevant equation) why replacing $\mathbf{W}$ with $\tilde{\mathbf{W}} = \mathbf{W}\mathbf{R}$, for an orthogonal matrix $\mathbf{R}$, does not change the likelihood of the model.

Solution. Since $\mathbf{R}\mathbf{R}^\top = \mathbf{I}$,

$$\tilde{\mathbf{W}}\tilde{\mathbf{W}}^\top = \mathbf{W}\mathbf{R}\mathbf{R}^\top\mathbf{W}^\top = \mathbf{W}\mathbf{W}^\top,$$

so $\text{Cov}[\mathbf{x}] = \tilde{\mathbf{W}}\tilde{\mathbf{W}}^\top + \boldsymbol{\Psi} = \mathbf{W}\mathbf{W}^\top + \boldsymbol{\Psi}$ is unchanged, and hence the marginal $p(\mathbf{x})$ – and therefore the likelihood – is identical. Geometrically, $\mathbf{R}$ corresponds to rotating $\mathbf{z}$, but $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ looks the same from every direction (isotropic), so the rotation has no effect.

- b) Give two of the proposed solutions to this identifiability problem, and briefly explain how each one resolves the ambiguity.

Solution. Possible answers (any two):

- **Orthonormal columns:** force $\mathbf{W}$ to have orthonormal columns (the PCA approach) – this makes the solution unique up to a permutation of the latent dimensions.
- **Lower-triangular $\mathbf{W}$:** force $\mathbf{W}$ to be lower triangular with positive diagonal entries, so the first visible feature depends only on the first factor, the second on the first two factors, etc.
- **Sparsity-promoting priors:** encourage entries of $\mathbf{W}$ to be (close to) zero via ℓ_1 regularisation, ARD, or spike-and-slab priors, giving interpretable (sparse) solutions.
- **Rotation methods** (e.g. orthogonal/oblique rotations such as varimax): fit the model first, then rotate the factors post-hoc to satisfy an interpretability criterion.

- c) Which of these solutions is used by PCA, and what additional ambiguity remains even after applying it?

Solution. PCA uses the orthonormal-columns solution. The remaining ambiguity is a permutation (and sign) ambiguity of the latent dimensions, which PCA resolves by ordering the components in decreasing order of eigenvalue (explained variance).

Question 5 – Probabilistic PCA (PPCA)

PPCA is a special case of factor analysis.

- a) What additional restrictions does PPCA place on $\mathbf{W}$ and $\mathbf{\Psi}$, compared to general FA?

Solution. PPCA requires $\mathbf{W}$ to have orthonormal columns and $\mathbf{\Psi} = \sigma^2 \mathbf{I}$ (isotropic noise), as opposed to general FA where $\mathbf{\Psi}$ is an arbitrary diagonal matrix and $\mathbf{W}$ is unconstrained.

- b) In the noise-free limit $\sigma^2 \rightarrow 0$, how does the MLE $\mathbf{W}_{\text{mle}}$ relate to the PCA solution?

Solution. In the limit $\sigma^2 = 0$, $\mathbf{W}_{\text{mle}} = \mathbf{U}_L \mathbf{L}_L^{1/2}$, where $\mathbf{U}_L$ are the top- L eigenvectors of $\mathbf{S}$ and $\mathbf{L}_L$ the corresponding eigenvalues. This is proportional to (i.e. a rescaled version of) the PCA solution $\mathbf{U}_L$.

- c) What is the interpretation of the MLE for σ^2 in terms of the discarded eigenvalues?

Solution. $\hat{\sigma}^2 = \frac{1}{D-L} \sum_{i=L+1}^D \lambda_i$, i.e. the average of the eigenvalues corresponding to the discarded dimensions – the average “distortion”/variance not captured by the retained L components. If $L = D$, $\hat{\sigma}^2 = 0$ since nothing is discarded.

- d) Write down the E step and M step of the EM algorithm for PPCA in the noise-free limit ($\sigma^2 = 0$).

Solution.

$$\tilde{\mathbf{Z}} = (\mathbf{W}^\top \mathbf{W})^{-1} \mathbf{W}^\top \tilde{\mathbf{X}} \quad (\text{E step}), \quad \mathbf{W} = \tilde{\mathbf{X}} \tilde{\mathbf{Z}}^\top (\tilde{\mathbf{Z}} \tilde{\mathbf{Z}}^\top)^{-1} \quad (\text{M step}).$$

- e) List two advantages of EM over standard eigenvector methods for fitting PCA.

Solution. Any two of: (1) faster for $N, D \gg L$, with cost $O(TLND)$ vs $O(\min(ND^2, DN^2))$ for naive eigendecomposition; (2) can be implemented online, updating $\mathbf{W}$ as data streams in; (3) can handle missing data simply; (4) can be extended to mixtures of PPCA/FA; (5) can be modified to variational EM / variational Bayes EM for more complex models.

Question 6 – Manifold learning and MDS

- a) What is the manifold hypothesis, and how does the digit-6 example illustrate the difference between intrinsic and ambient dimensionality?

Solution. The manifold hypothesis states that most naturally occurring high-dimensional datasets lie (approximately) on a low-dimensional manifold embedded in the high-dimensional ambient space. For the digit-6 example, each image is a vector in a 3,648-dimensional space (ambient dimensionality D), but rotating the same digit through 360 different angles and applying PCA reveals that the variation lies on a curved 2d manifold – the intrinsic dimensionality $d = 2$ is much smaller than $D = 3,648$.

- b) What is the goal of multidimensional scaling (MDS), and what quantity does classical MDS try to preserve?

Solution. MDS aims to find low-dimensional embeddings $\{\mathbf{z}_i\}$ such that pairwise distances/dissimilarities between the embeddings match a given set of pairwise dissimilarities d_{ij} as closely as possible. Classical MDS specifically tries to preserve the (centered) Gram/similarity matrix $\hat{\mathbf{K}}$, i.e. the inner products between centered data points, via the strain objective $\|\hat{\mathbf{K}} - \hat{\mathbf{Z}}\hat{\mathbf{Z}}^\top\|_F^2$.

- c) What is the difference between metric MDS and non-metric MDS? In particular, what does non-metric MDS try to match instead of distances?

Solution. Metric MDS minimises the squared difference between the given dissimilarities d_{ij} and the embedding distances $\hat{d}_{ij} = \|\mathbf{z}_i - \mathbf{z}_j\|$ directly – it is a different (generally non-equivalent) objective from classical MDS even when d_{ij} are Euclidean. Non-metric MDS instead tries to match the *ranking* (ordering) of the dissimilarities, via a monotonic transformation $f(d)$ from distances to ranks, optimised jointly with the embedding.

Question 7 – t-SNE and Kernel PCA

- a) In SNE/t-SNE, what do the quantities $p_{j|i}$ and $q_{j|i}$ represent, and what objective is used to match them?

Solution. $p_{j|i}$ is the probability that point i would pick point j as its neighbour, based on Euclidean distances in the high-dimensional (input) space (via a Gaussian kernel). $q_{j|i}$ is the analogous probability computed from distances between the low-dimensional embeddings $\mathbf{z}_i, \mathbf{z}_j$. SNE matches the two distributions by minimising the (sum of) KL divergence(s) between P and Q .

- b) What is the “crowding problem”, and how does t-SNE address it?

Solution. The crowding problem is the tendency of SNE-type methods to squeeze points that are relatively far apart in high-dimensional space close together in the low-dimensional embedding, because Gaussian-based similarities decay too quickly with distance. t-SNE addresses this by using a heavy-tailed Student- t (with $\nu = 1$, i.e. Cauchy) distribution for q_{ij} in the low-dimensional space, which allows moderately distant points in the embedding to still represent moderate similarity, reducing unwanted crowding.

- c) What is the key idea behind kernel PCA, and what types of nonlinear structure can it capture that linear PCA cannot?

Solution. Kernel PCA performs PCA implicitly in a (possibly infinite-dimensional) feature space $\phi(\mathbf{x})$ by replacing inner products with a kernel function $\mathcal{K}(\mathbf{x}_i, \mathbf{x}_j)$ (the kernel trick), using the eigendecomposition of the Gram matrix $\mathbf{K}$. This allows it to capture nonlinear relationships/curved structure in the data (e.g. clusters separated by nonlinear boundaries) that a linear projection (standard PCA) cannot represent, since standard PCA is limited to linear combinations of the original features.